

GPA ANALYZER & PREDICTOR

A Python tool for calculating GPA, CGPA, and academic trend projections



The Problem

Students track results across many semesters but rarely see the full picture: how each semester affects overall CGPA, what a future semester could do to it, and what grades are actually needed to hit a target.

Goal

Turn a raw results CSV into a clear picture of academic performance and trend.

Audience

Students and registry units who need fast, accurate GPA/CGPA tracking.

Approach

Clean data handling, tested calculation logic, and simple 'what-if' projections.

Core Features

1

Load & validate data

Reads results.csv, drops bad rows (invalid grade, bad units) without crashing, reports what was dropped.

2

Per-semester GPA

Weighted by course units, on the 5-point scale (A=5 ... F=0).

3

Cumulative CGPA

Tracked across every semester, updated as new data comes in.

4

Trend chart

GPA and CGPA plotted together across semesters, in chronological order.

5

Scenario tool

"If I get X next semester" — projects CGPA from a hypothetical semester.

6

Target-CGPA advice

Calculates the average grade point needed to hit a target CGPA, or says if it's not reachable.

Tech Stack & Structure

Python 3.11+

Pandas

Matplotlib

Jupyter Notebook

Pytest

Scikit-learn (optional)

Project structure

- gpa_core.py — pure calculation & validation functions
- gpa_analyzer.ipynb — main notebook interface
- test_gpa_core.py — 8 pytest unit tests
- sample_results.csv — sample dataset
- README.md — setup & usage documentation

Tested Before Submission

8 automated pytest cases, plus a manual walkthrough of every feature.

```
pytest -v test_gpa_core.py
```

```
test_gpa_single_semester PASSED
test_cgpa_across_semesters PASSED
test_invalid_grade_dropped PASSED
test_zero_units_row_skipped PASSED
test_scenario_projection_matches_manual_calc PASSED
test_target_cgpa_achievable PASSED
test_target_cgpa_unreachable PASSED
test_empty_dataframe PASSED
```

Manual checklist

- ✓ Restart + Run All, top to bottom, clean
- ✓ Chart order confirmed chronological
- ✓ Scenario tool moves CGPA the right direction
- ✓ Broken CSV handled without crashing
- ✓ Unreachable target flagged, not miscalculated

GPA Analyzer & Predictor

Clean data handling, tested calculation logic, honest projections.

github.com/sQuiDbOb18